

Time Series Analysis of Stock Prices

Objective

The objective of this project is to analyze stock price data over time and identify trends, seasonal patterns, and future price behavior.

Dataset Description

The dataset contains historical stock market data including:

- Date
- Open Price
- High Price
- Low Price
- Close Price
- Volume

Trend Analysis

The stock closing price was analyzed over time. The trend graph shows how the stock price changed throughout the observed period.

A 30-day moving average was used to smooth short-term fluctuations and highlight long-term trends.

Seasonality Detection

Monthly average stock prices were calculated to identify recurring patterns. This helps determine whether specific months tend to perform better or worse than others.

Forecasting

A simple forecasting model was applied to estimate future stock prices based on historical trends.

Key Insights

1. The stock exhibited an overall upward/downward trend during the analysis period.
2. Moving averages helped identify long-term price movement.
3. Monthly analysis revealed seasonal variations in stock performance.
4. Forecasted values provide an estimate of future price direction.

Conclusion

Time series analysis helps investors understand stock behavior, identify trends, and make informed investment decisions based on historical data patterns.

Screenshots:

